

Introduction to Simulation Model

1EDN71x6U

Targeted simulator: SIMetrix

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Gate driver model specifications

1. General considerations

The 1EDN71x6U simulation model is written in PSpice. The model development has been performed according to the datasheet.

2. Functionality

Model performances:

- Static Electrical Characteristics and Dynamic Electrical Characteristics are modeled with the typical values from the datasheet.
- Temperature effects are modeled

Implemented features:

- Switching Characteristics
 - Propagation delay
 - Peak/mean currents
 - Output resistance
- Under voltage lockout
- Current consumption

3. Descriptions of ports/pins

Port name	Port type
IN+	Connected to PWM output controller via 47 kohm or 75 kohm resistor
IN-	Connected to controller ground via 47 kohm or 75 kohm resistor
VSS	Return path for VDD and thermal dissipation pad
VDD	Gate drive supply
BST	Bootstrap diode anode connection point, when used as low-side driver in a half-bridge configuration
OUT_SNK	Low-impedance gate pull-down to VOFF (including active Miller clamp)
OUT_SRC	Low-impedance gate pull-up to VDD
TAMB	Ambient temperature
TJ	Junction temperature

4. Simulation options

Maximum time step of 1ns have to be used for correct timing parameters

To be able to get the correct timing parameters of the model, the maximum time step has to be set to 1ns or lower. Please note that a lower maximum time step increases the simulation time, but leads to more accurate results.

In order to choose the simulation maximum time step in SIMetrix, please go to:
Simulator -> Choose analysis -> Transient -> Advanced options.

